

Shane Miller

Applied AI Field Engineer & Software Developer

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PROFESSIONAL SUMMARY

Software developer with 12+ years of experience building full-stack applications and enterprise systems. As an Applied AI Field Engineer at Trilon Group, I conduct onsite deep dives with engineering firms and rapidly turn complex workflows into working software: deterministic calculation and document-processing tools, human-in-the-loop AI workflows, and the data architecture underneath them.

PROFESSIONAL EXPERIENCE

Applied AI Field Engineer

June 2026 – Present

Trilon Group — Hybrid

Work within a cross-functional team conducting onsite deep dives with engineering firms, documenting workflows, data sources, and calculations, then identifying where automation, AI, or purpose-built software delivers measurable value. Rapidly prototype full-stack applications: deterministic calculation, validation, and document cross-referencing tools, and human-in-the-loop AI workflows with reviewable results, citations, and traceability. Delivered environmental workflow automation over GIS and regulated modeling tools, civil engineering calculations for stormwater and hydrology, survey-data processing and coordinate transformation, and grant application scoring software. Write structured, reusable code and phased implementation plans so successful prototypes have a clear path to production.

Senior Engineer Level II

April 2025 – June 2026

Data Center Inc. — Remote

Delivered full-stack C#/NET features across ACH processing, batch systems, transaction flows, reconciliation, and reporting on a core banking platform for community financial institutions. Diagnosed production incidents involving live financial data and downstream systems. Improved reliability through performance and static code analysis, partnering with product, QA, business analysts, and support to deliver secure releases in a regulated environment.

Founder & Lead Engineer

April 2024 – Present

U-Mail.ai — Boulder, CO

Built an email productivity platform across web and mobile with secure account connections, deterministic message prioritization, and AI-assisted composition. Developed Node.js and Express APIs with Gmail, Google Calendar, Firebase, OAuth 2.0, Google Meet, Zoom, and secure token integrations. Designed reusable services and multi-tenant architecture for consumer and B2B deployments, managing releases, observability, privacy controls, and reliability.

Founder & Lead Engineer

September 2024 – Present

ApplyPro.ai — Boulder, CO

Built and launched an AI-powered resume platform, owning product design, frontend, backend, payments, analytics, and deployment. Developed a server-rendered Next.js application with App Router, TypeScript, and document workflows. Integrated Stripe subscriptions, Mixpanel, and Vercel CI/CD, scaling to 400+ active users through rapid iteration and customer feedback.

Senior Software Engineer

February 2018 – February 2024

S&P Global — Boulder, CO

Resolved high-impact production incidents affecting real-time financial data and institutional clients across Vue, React, JavaScript, Node.js, and C#/.NET. Owned SAML, SSO, and OAuth integrations across distributed products. Optimized high-volume ingestion and cross-referencing pipelines for global exchange and vendor feeds, coordinating with data providers, infrastructure, and product teams to protect uptime and meet service-level commitments.

Web Developer — Client-Facing Team

August 2014 – 2016

S&P Global — Boulder, CO

Built and maintained high-traffic financial web applications and live-streaming interfaces for multinational institutional clients. Delivered client-specific enhancements across mature codebases and worked directly with global customers to troubleshoot production issues under tight deadlines.

TECHNOLOGY & TOOLS

Frontend: Next.js, React, TypeScript, Tailwind CSS

Backend & Automation: Python, APIs, Server-Side Processing

AI & Machine Learning: Large Language Models (LLMs), Retrieval-Augmented Generation (RAG), AI Agents, Vector Search, Model Evaluation, Model Context Protocol (MCP)

Data & Storage: Azure Blob Storage, SQL, JSON, Excel, CSV, Metadata Models, Versioning & Audit Trails

Identity & Integrations: Microsoft Entra ID, Microsoft Graph, SharePoint, Egnyte

Engineering & Geospatial: ArcGIS, GIS Data, CAD / DXF, Coordinate Systems, Engineering Calculation Engines

Development: Git, GitHub, VS Code, Monorepo Architecture, CI/CD

Delivery & Documentation: Jira, Kanban, User Stories & Acceptance Criteria, Technical Audits, Phased Implementation, Markdown, Architecture & Implementation Guides, Demonstrations & Technical Handoffs

Additional Experience: C# / .NET, Node.js / Express, SQL Server, PostgreSQL, React Native, Vue.js, SAML / SSO / OAuth 2.0, Firebase, Google Cloud, Vercel

EDUCATION

Certificate in Mid-Tier Development · IHS Markit / Dev-U · October–November 2014

Web Development Immersive · General Assembly · December 2013–July 2014